

Route-F P54: Goldstone Infrared Audit, Scalar Momentum Dependence and Background Control

P54PQ-v2 four-dimensional mainline

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Abstract

We retain the P54PQ-v2 action and first audit its frozen scalar point. All 34 Goldstone hard mass shifts cancel against the same finite fixed-VEV counterterms by the full orbit Ward identity. The excluded soft scalar tadpoles vanish, but their radial potential Hessian has a nonzero logarithmic infrared singularity. We compute its actual Gram coefficient and the complete one-loop scalar momentum dependence on the three-dimensional radial slice. The hard scalar kinetic correction has a largest generalized eigenvalue 2.61, independently exposing a failure of a small-insertion expansion. We then analytically rule out an entire uniformly weakened bosonic parameter ray at fixed radii. Four less-hierarchical scalar engineering points are tested separately; none passes both radial positivity and perturbative-control criteria. No physical benchmark, flavor fit, full gauged pole kernel, or complete finite matching is claimed. The original inputs are not overwritten.

1 Fixed action, conventions and acceptance boundary

The scalar content is real 54, complex 126, complex 10 and complex S , with three chiral 16_F families. The original radial point is

$$r_0 = (w, \sigma, v_s) = (1, 0.1265, 0.25)\omega, \quad \mu_0 = g_U \omega = 0.5626028899853225\omega. \quad (1)$$

The same $\overline{\text{DR}}/\overline{\text{MS}}$ background-field Landau convention and common fixed-VEV mass counterterms of P-REN1 are used. The 328-dimensional real canonical scalar Hessian has 38 soft tree modes: 33 gauge Goldstones, one physical PQ mode, and four components of the tuned light doublet. There are 290 positive hard scalar modes and 33 massive gauge bosons. The 12 unbroken vectors remain massless along this radial slice. Their vanishing potential contribution does not eliminate all momentum-dependent gauge diagrams.

Let R embed r in the canonical field X , so

$$G = R^T R = \text{diag}(12/5, 2, 1), \quad \mathbf{H} = V_0''(X), \quad T_r = \partial_r \mathbf{H}, \quad U_{rs} = \partial_r \partial_s \mathbf{H}. \quad (2)$$

All these are obtained from the actual polynomial action. Its quartic degree makes $\mathbf{H}$ quadratic, hence cached radial jets determine its exact radial dependence without a new Hessian evaluation or a fitted mass formula. Floating-point residuals only test these identities.

Goldstone resummation and a finite physical scalar mass are distinct questions. The former does not by itself render the second derivative of a zero-momentum potential finite; external-momentum self-energies are needed [1]. Hard/soft separation also matters in organizing the resummation [2]. We apply these distinctions rather than introduce a positive Goldstone mass as an adjustable input.

2 Common tadpoles fix the Goldstone shift

For any antisymmetric generator T of a scalar symmetry and invariant function V , differentiating $\nabla V \cdot TX = 0$ gives

$$V''(X)TX = T\nabla V(X). \quad (3)$$

This applies separately to the locally separated hard spectral functional $V_{1,h}$ and to its invariant finite counterterm. On the radial background, $\nabla V_{1,h} = RG^{-1}t_h$. The common tadpole condition is $t_h + \partial_r V_{CT} = 0$. Consequently

$$(V_{1,h}'' + V_{CT}'')TX = 0. \quad (4)$$

Collecting the 45 gauge generators and the PQ generator into the redundant orbit matrix $B = (T_a X)$, Eq. (3) determines the whole Goldstone block as

$$B^T V_{1,h}'' B = B^T (T_a \nabla V_{1,h})_a. \quad (5)$$

Its normalized 34-dimensional hard block has eigenvalues between $-2.7348720\omega^2$ and $-0.0342377\omega^2$. The common counterterm cancels it; the matrix residual is $7.8 \times 10^{-15}\omega^2$. The actual tree cubic vertices independently obey the differentiated identity

$$Q^T T_r (T_a X) = Q^T T_a H R_r, \quad (6)$$

where Q spans the full tree kernel. This follows by differentiating Eq. (3) for V_0 and using $Q^T H = 0$. All three radial derivatives are checked directly.

In a reorganized propagator the Goldstone shift must include its counterterm in this convention. Resumming the bare hard shift alone, or replacing it by a freely chosen positive regulator, would not describe the same prescribed point. Equation (4) is a one-loop Ward result, not a completed two-loop resummation calculation.

3 The actual soft infrared singularity

3.1 Gaussian determinant and the soft graph coefficient

For real scalar fluctuations, Gaussian integration gives

$$\Gamma_1^S = \frac{1}{2} \text{Tr} \log(-\partial_E^2 + H), \quad V_1^S = \frac{1}{64\pi^2} \text{tr} \left[H^2 \left(\log \frac{H}{\mu_0^2} - \frac{3}{2} \right) \right]. \quad (7)$$

Let Q span the 38 soft modes and let P_h span their massive complement. Set

$$A_r = Q^T T_r Q, \quad (8)$$

$$\mathcal{B}_{rs} = \frac{\text{tr}(A_r A_s)}{32\pi^2}, \quad (9)$$

$$\mathcal{E}_{rs} = \frac{1}{32\pi^2} \left[\text{tr}(Q^T U_{rs} Q) - 2 \sum_{a \in s, i \in h} \frac{(T_r)_{ai} (T_s)_{ia}}{m_i^2} \right]. \quad (10)$$

The last term is essential: second-order eigenvalue perturbation contains mixing with the hard complement. Freezing the soft projector through second order would omit it.

Regulate the soft eigenvalue branches by adding a common mass squared $\epsilon > 0$, without changing the hard eigenvalues. Their tadpole and Hessian at the reference point are

$$t_r^s(\epsilon) = \frac{\epsilon}{32\pi^2} \left(\log \frac{\epsilon}{\mu_0^2} - 1 \right) \text{tr} A_r \longrightarrow 0, \quad (11)$$

$$H_{rs}^s(\epsilon) = \mathcal{B}_{rs} \log \frac{\epsilon}{\mu_0^2} + \epsilon \left(\log \frac{\epsilon}{\mu_0^2} - 1 \right) \mathcal{E}_{rs}. \quad (12)$$

These follow from $f'(\epsilon) = 2\epsilon(\log(\epsilon/\mu_0^2) - 1)$, $f''(\epsilon) = 2\log(\epsilon/\mu_0^2)$ and the reduced second eigenvalue derivative. Direct spectral finite differences independently verify Eq. (12).

For a real radial vector v ,

$$v^T \mathcal{B} v = \frac{1}{32\pi^2} \left\| \sum_r v_r A_r \right\|_F^2 \geq 0. \quad (13)$$

Thus the leading infrared logarithm is negative semidefinite as $\epsilon \rightarrow 0$. It is not a finite positive repair of a negative hard curvature. In the declared radial coordinates, the actual matrix is

$$\frac{\mathcal{B}}{\omega^2} \simeq \begin{pmatrix} 0.60561772 & 0.00943837 & 0.00589905 \\ 0.00943837 & 0.00048543 & 0.00020861 \\ 0.00589905 & 0.00020861 & 0.00058130 \end{pmatrix}. \quad (14)$$

The smallness of its projection on one negative direction does not mean that the whole matrix is small.

For the normalized negative witness q retained from P-REN1, $q^T G q = 1$, its decomposition is

Soft sector	$q^T \mathcal{B} q / \omega^2$
Gauge Goldstones	3.1096973×10^{-4}
Physical PQ	1.8983181×10^{-6}
Light Higgs	2.6290000×10^{-7}
Total	3.1313095×10^{-4}

Mixed sector contributions vanish for these radial vertices, as verified with the actual projectors. At $\epsilon/\omega^2 = 10^{-4}$ and 10^{-10} , respectively, $q^T H^s q / \omega^2$ is approximately -0.00255 and -0.00685 . This is regulator dependence, not a pair of physical mass predictions. The zero-momentum full potential Hessian cannot be promoted to a pole mass by merely restoring the missing modes.

4 Infrared cancellation at nonzero momentum

4.1 Complete one-loop scalar term on the radial slice

Define the Euclidean finite bubble with $s = p_E^2 > 0$:

$$L(a, b; s) = \int_0^1 \log \frac{(1-x)a + xb + x(1-x)s}{\mu_0^2} dx. \quad (15)$$

It is the negative of the usual finite B_0 in this convention. Differentiating the scalar functional determinant twice gives

$$\Pi_{rs}^S(s) = \frac{1}{32\pi^2} \left[\sum_i (U_{rs})_{ii} m_i^2 \left(\log \frac{m_i^2}{\mu_0^2} - 1 \right) + \sum_{ij} (T_r)_{ij} (T_s)_{ji} L(m_i^2, m_j^2; s) \right]. \quad (16)$$

The first term is the seagull and the second the two-cubic-vertex bubble; ordered real-field indices account for the symmetry factor. The sums include all 328 scalars, including mixed hard/soft pairs. At $s = 0$, removing only the soft-soft bubble reproduces the old scalar hard Fréchet Hessian to relative residual 1.4×10^{-16} .

For two massless lines,

$$L(0, 0; s) = \log \frac{s}{\mu_0^2} - 2. \quad (17)$$

For one massive line m^2 and $z = s/m^2$,

$$L(0, m^2; s) = \log \frac{m^2}{\mu_0^2} - 2 + (1 + z^{-1}) \log(1 + z). \quad (18)$$

Independent Feynman-parameter integrals check both finite constants. The logarithmic regulator cancels explicitly:

$$\begin{aligned} H^s(\epsilon) + \mathcal{B} \left[L(\epsilon, \epsilon; s) - \log \frac{\epsilon}{\mu_0^2} \right] \\ \longrightarrow \mathcal{B} \left(\log \frac{s}{\mu_0^2} - 2 \right). \end{aligned} \quad (19)$$

At fixed $s > 0$ this is finite. It does not license taking a physical mass from the $s = 0$ potential curvature.

For at least one massive line,

$$L(a, b; s) - L(a, b; 0) = \int_0^1 \log \left[1 + \frac{x(1-x)s}{(1-x)a + xb} \right] dx \geq 0. \quad (20)$$

Combined with the vertex Gram form, the hard scalar momentum difference $\Delta\Pi_h^S(s)$ is positive semidefinite. By contrast, the massless contribution is negative semidefinite for $0 < s \leq e^2\mu_0^2$ in this MS prescription.

The displayed diagnostic kernel is

$$K_{\text{diag}}(s) = sG + H_{r,B} + \Delta\Pi_h^S(s) + \mathcal{B}[\log(s/\mu_0^2) - 2], \quad (21)$$

where $H_{r,B}$ is the old bosonic hard-plus-counterterm radial block. The vector contribution is still evaluated at zero momentum.

s/ω^2	.001	.01	.05	.1	.5	1
$\lambda_{\min}(K_{\text{diag}}, G)/\omega^2$	-.75843	-.72498	-.58421	-.41369	.59623	1.09859

These are kinematic probes, not a parameter scan or physical pole locations. Missing vector/Goldstone/ghost mixed diagrams, fermion momentum maps and Nielsen consistency prevent interpreting a zero of this partial kernel as a physical tachyon or a stable-particle pole.

4.2 A separate quantitative failure of small-insertion control

The finite hard scalar kinetic matrix is

$$(\Delta Z_h^S)_{rs} = \frac{1}{32\pi^2} \sum_{(i,j) \notin (s,s)} (T_r)_{ij} (T_s)_{ji} \int_0^1 \frac{x(1-x) dx}{(1-x)m_i^2 + xm_j^2}. \quad (22)$$

It is positive semidefinite and its generalized eigenvalues relative to G are

$$(0.00900848, 0.27686405, 2.61031989). \quad (23)$$

Finite differences of the actual momentum kernel verify this derivative. The soft-soft contribution is nonlocal and is not included in Eq. (22). The largest hard correction already exceeds the tree metric, so the naive small-insertion expansion fails this necessary engineering check at the frozen point.

A finite canonical field transformation can absorb a positive metric, but is not by itself a bound on omitted loop orders. Moreover, an invertible congruence of a fixed curvature matrix preserves its inertia. It cannot turn its negative direction positive simply by renormalizing the fields. These statements are not an all-orders exclusion.

5 An entire simple repair ray fails analytically

Uniformly multiply every existing scalar coefficient by $z > 0$, including the dependent masses and the cubic coefficient, keeping r_0, g_U, μ_0 fixed. Then $V_0 \mapsto zV_0$ exactly: stationarity, the tree kernel and tree inertia are unchanged, while H, T_r, U_{rs} each scale by z . The common fixed-VEV scalar contribution therefore obeys

$$H_h(z) = zH_0 + z^2[S + \log z L_S] + V, \quad \Delta Z_h^S(z) = z\Delta Z_h^S(1). \quad (24)$$

Here S includes the scalar loop and its own share of the common counterterm, V the vector loop and its share, and L_S the coefficient of the scalar logarithm including its tadpole conversion. Direct rescaled Fréchet calculations check Eq. (24); the following bound is not inferred from the sampled points.

Proposition 1. *For every $0 < z \leq 1$, the bosonic hard fixed-VEV radial block on this ray has a negative direction.*

Proof. In the sigma coordinate the actual coefficients, in units of ω^2 , are

$$\begin{aligned} a &= (H_0)_{\sigma\sigma} = 0.02880405, & b &= -S_{\sigma\sigma} = 1.54130008, \\ (L_S)_{\sigma\sigma} &= 7.61695881, & c &= V_{\sigma\sigma} = -0.00402215. \end{aligned} \quad (25)$$

Since $\log z \leq 0$, one has

$$(H_h(z))_{\sigma\sigma}/\omega^2 \leq az - bz^2 + c \leq c + \frac{a^2}{4b} = -0.00388757 < 0. \quad (26)$$

The normalized radial direction is $e_\sigma/\sqrt{2}$, and remains negative for the entire interval. $\square$

The theorem concerns this bosonic ray only. It does not exclude nonuniform scalar choices, different radial hierarchies, or fermionic corrections at new scalar points. It shows why merely reducing all scalar couplings is not enough in the fixed gauge/scale convention.

6 Bounded reselection without fitting observations

After the original control failure, four coarse engineering points were tested with $(z, \sigma/\omega) = (.03, .3), (.03, .5), (.1, .3), (.1, .5)$ and $w/\omega = 1$, $v_s/\omega = .25$. All existing scalar coefficients are first scaled by z . The dependent masses are then re-solved from

$$A(r) \delta p = -\partial_r V_0(r; zp), \quad A(r) = -\text{diag}(12w/5, \sigma, v_s). \quad (27)$$

The tree ξ_{02} condition is reimposed at the first physical doublet crossing. Every point has 34 symmetry zero modes plus precisely one complex light doublet, checked with the actual SM Casimirs.

The polynomial identity

$$\mathbf{H}(r_0 + \Delta\sigma e_\sigma; p) = \mathbf{H}_0 + \Delta\sigma T_\sigma + \frac{1}{2}(\Delta\sigma)^2 U_{\sigma\sigma} \quad (28)$$

and the exact mass-parameter operator give the new full ambient tree Hessian without a fresh large tensor differentiation. New tadpoles, common counterterms and radial hard/kinetic matrices are then evaluated from each point's own spectrum, not borrowed from the old point.

We demand radial hard positivity and test two dimensionless control quantities, with a declared engineering margin 0.3:

$$\rho_H = \max |\text{spec}(H_h - H_0, H_0)|, \quad \rho_Z = \max \text{spec}(\Delta Z_h^S, G). \quad (29)$$

This margin is not a theorem of perturbative convergence; importantly, the rejected positive point even fails the weaker $\rho_H < 1$ test.

z	σ/ω	$\lambda_{\min}(H_h, G)/\omega^2$	ρ_H	ρ_Z
.03	.3	-.040838	18.8648	.09010
.03	.5	-.044064	7.9561	.09296
.1	.3	-.198595	27.0000	.30032
.1	.5	.009714	1.4323	.30988

Thus none is accepted. The last point illustrates why positivity alone cannot certify a controlled background. This is not a no-go theorem for all less-hierarchical points. Full parameter cards, finite tadpole shifts, tree classifications and rejection evidence are retained in the reselection JSON. No measured mass or flavor datum enters. Changing the hierarchy invalidates reuse of the old two-loop unification/threshold solution; g_U was held only as a reference coupling for these engineering probes.

7 Status and next mathematical tasks

The IR verifier passes 65/65 checks and the bounded reselection verifier passes 36/36. Passing includes correctly rejecting physical promotion. Both reuse ten cached Hessians and require no new Hessian evaluations. The new artifacts are `code/verify_p54_goldstone_ir.py` and `code/verify_p54_scalar_reselection.py`, with matching JSON/Markdown reports under `output/`; sources and hashes are recorded.

The useful next options are now more specific:

1. Decompose the actual scalar-invariant contributions to ρ_H and ρ_Z , especially the independent 126 quartics, before nonuniform parameter adjustment. Bare couplings smaller than one and positive selected masses are not sufficient control criteria. Do not fine-tune against the displayed Euclidean subset.
2. Complete the same-prescription gauge/Goldstone/ghost and fermion momentum maps with Nielsen consistency if deciding the physical fate of the original point. A true pole/background claim cannot follow from a scalar-only repair or the partial kernel in Eq. (21).
3. After finding a controlled common scalar point, recompute its full nonradial response, one-loop light state, upper PS saddle and threshold/running system. Complete Wilson/box, lower finite matching and finite seesaw remain independent unfinished calculations. None is promoted to a physical fit by this audit.

References

- [1] J. Braathen and M. D. Goodsell, Avoiding the Goldstone Boson Catastrophe in general renormalisable field theories at two loops, especially the distinction between tadpole resummation and momentum-regulated masses, [arXiv:1609.06977v2](#). The two-loop mass results in that work specialize to the gaugeless limit; they are not a completed P54 gauge-sector calculation.
- [2] J. R. Espinosa and T. Konstandin, Resummation of Goldstone Infrared Divergences: A Proof to All Orders, [arXiv:1712.08068](#).